Terraform Project

Project Overview

This project uses **Terraform** to provision and manage AWS infrastructure in an automated way. The setup creates a complete environment including EC2 instances templates, security groups, autoscalling and supporting AWS resources..

This project is suitable for **learning Terraform**, **AWS infrastructure automation**, and **DevOps practices**.

In this project implement the following architecture via **terraform**.

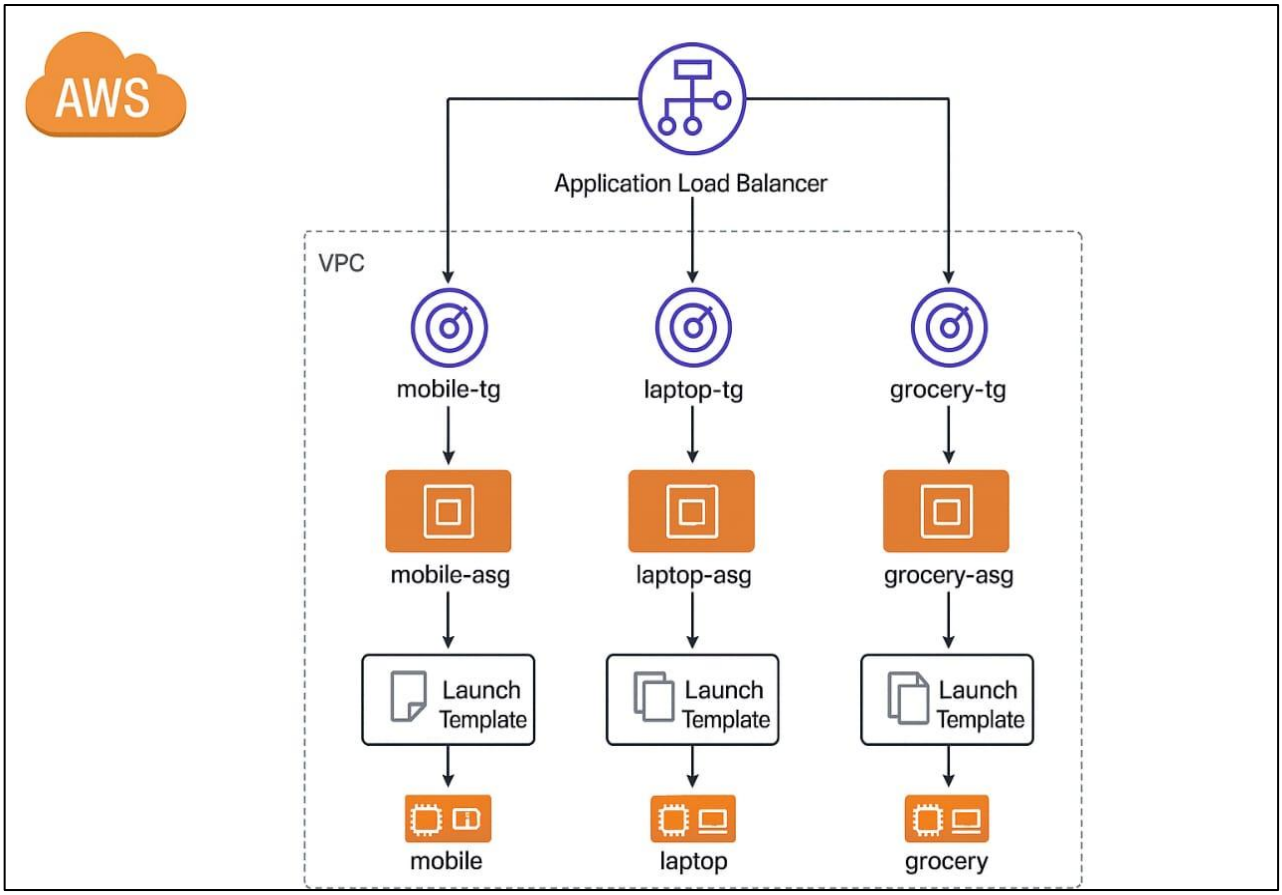


Fig.Architecture Diagram

Diagram description:

1. Application Load Balancer
2. VPC
3. Mobile target group, laptop target group, and grocery target group.
4. Mobile, laptop, grocery autoscalling group.
5. Launch Template for mobile, laptop, and grocery.

Technologies Used

- **Terraform** (Infrastructure as Code)
- Amazon Web Services (AWS)
  - EC2
  - VPC

- o Security Groups
- o Subnet
- o Internet Gateway
- o Route table
- o Application load balancer

❁ Prerequisites

Before running this project, make sure you have:

- An AWS account
- **Terraform installed** (v1.x recommended)
- AWS CLI installed and configured

🗁 Project Structure.

```
├── main.tf              # Main Terraform resources definition
├── vars.tf              # Variable declarations
├── outputs.tf           # show outputs after execution
├── terraform.tfvars     # Variable values
├── awsprofile.tf        # AWS provider & profile configuration
└── README.md            # Project documentation
```

Project execution :

1. Initialize Terraform
   Downloads required providers and initializes backend:
   terraform init

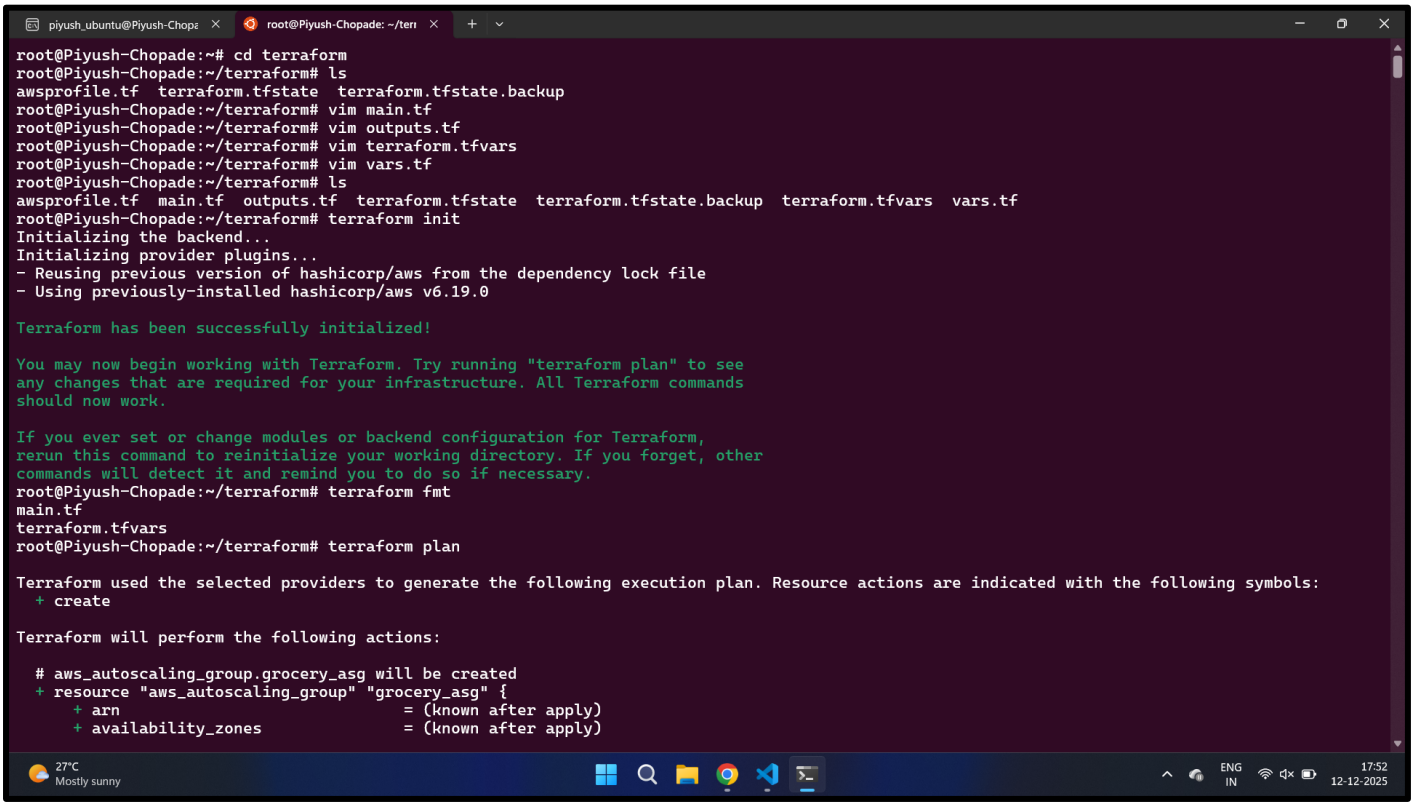
2. Validate Terraform Configuration
   Checks syntax and configuration correctness:
   terraform validate

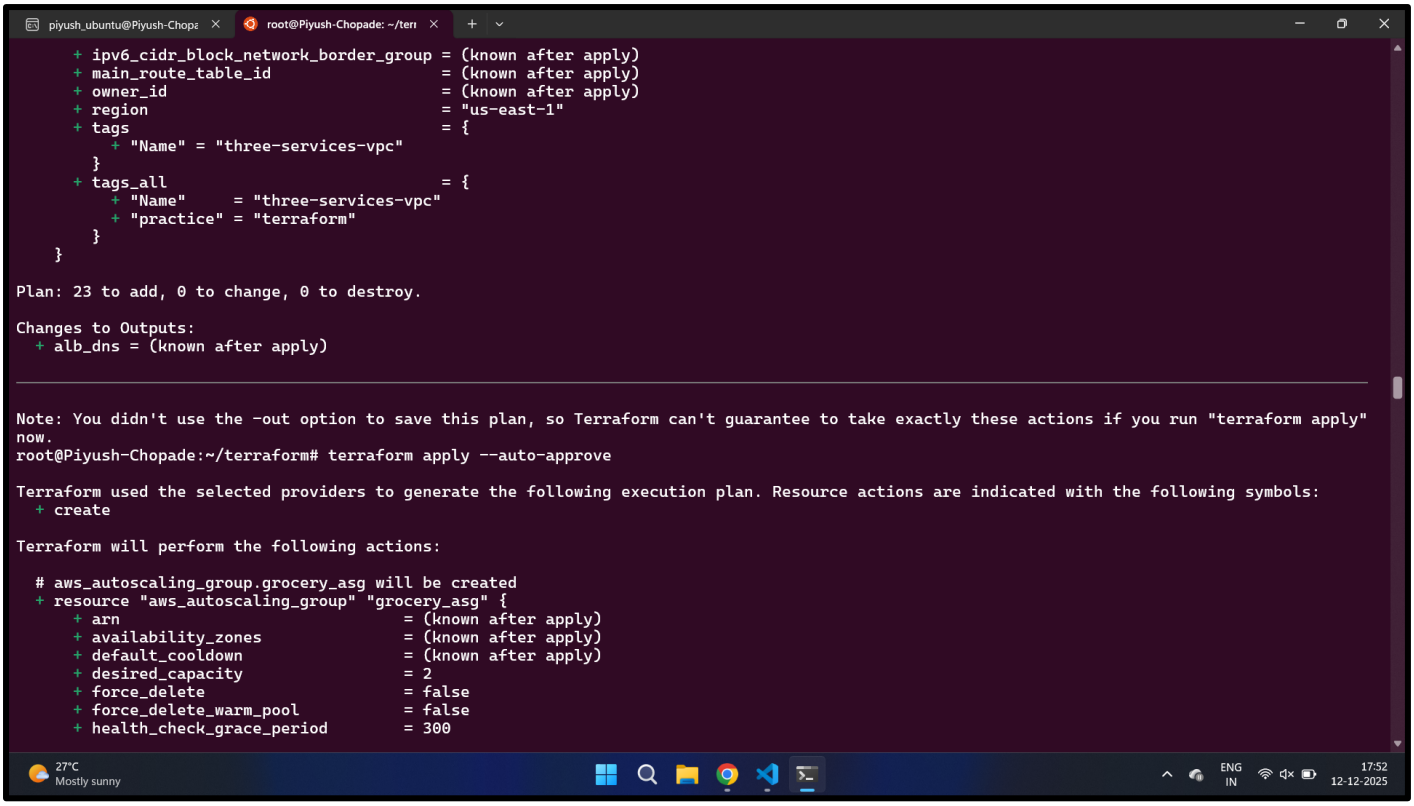
3. Preview Infrastructure Changes:
   Shows what resources will be created:
   terraform plan

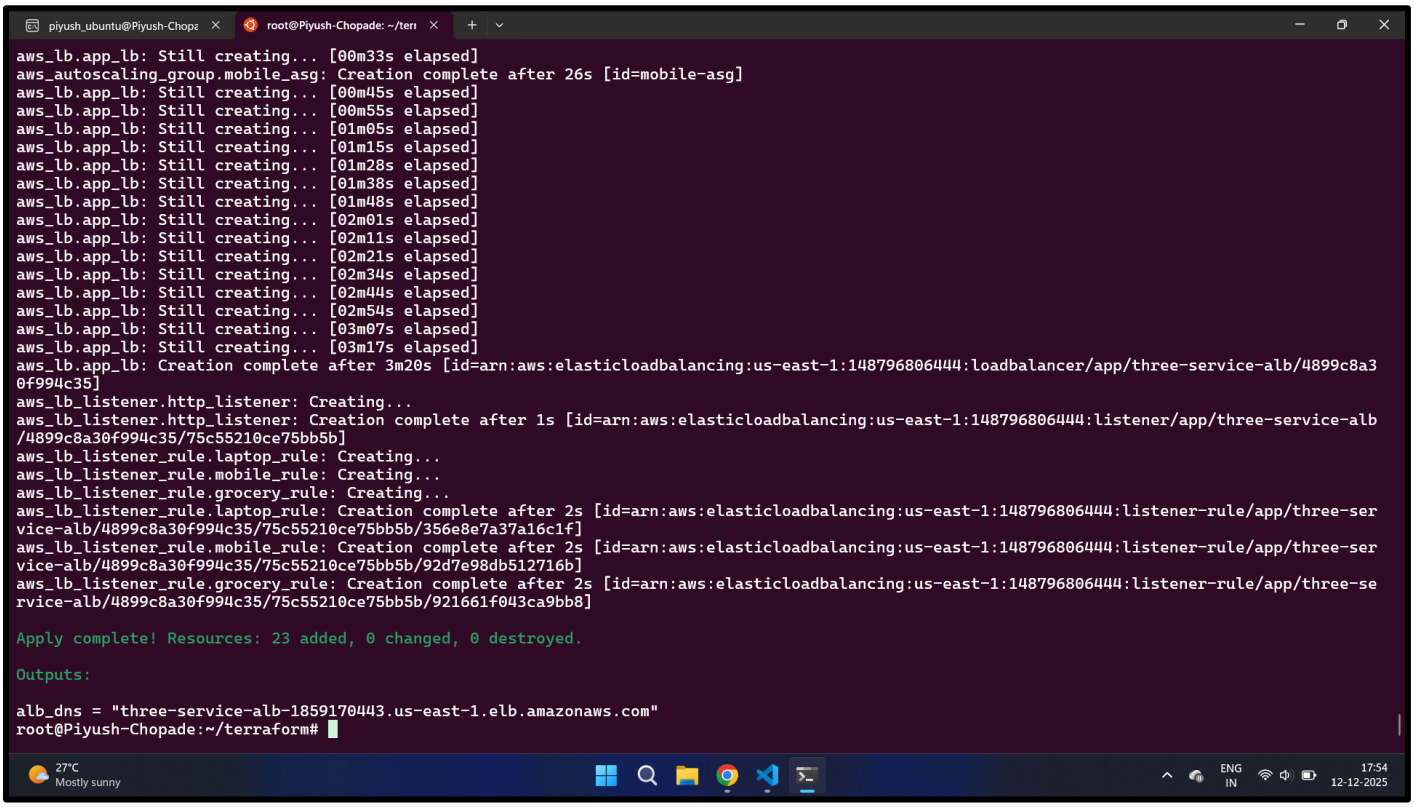
4. Apply Terraform Configuration
   Creates AWS infrastructure:
   terraform apply--auto-approve

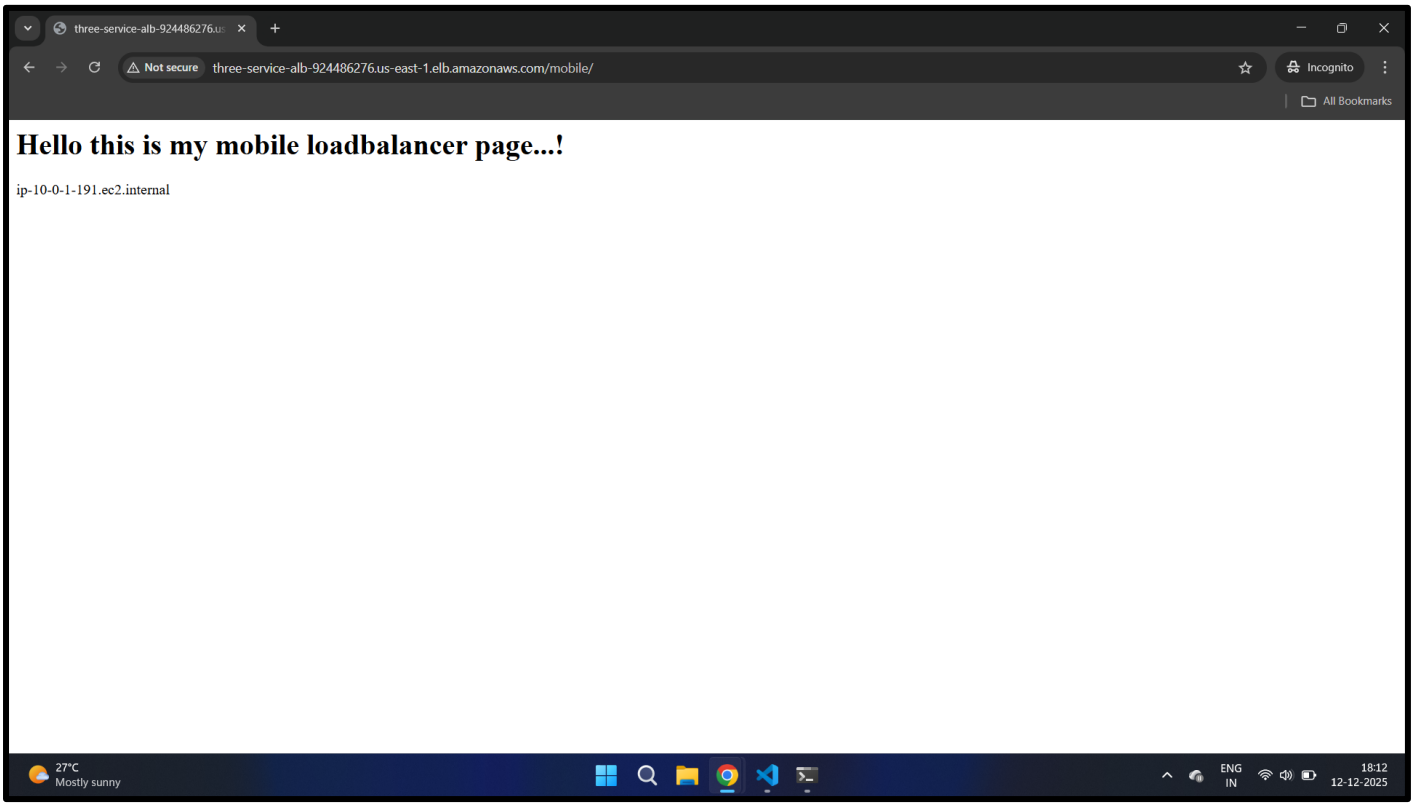
5. Apply Destroy Configuration
   destroy AWS infrastructure:
   terraform destroy--auto-approve

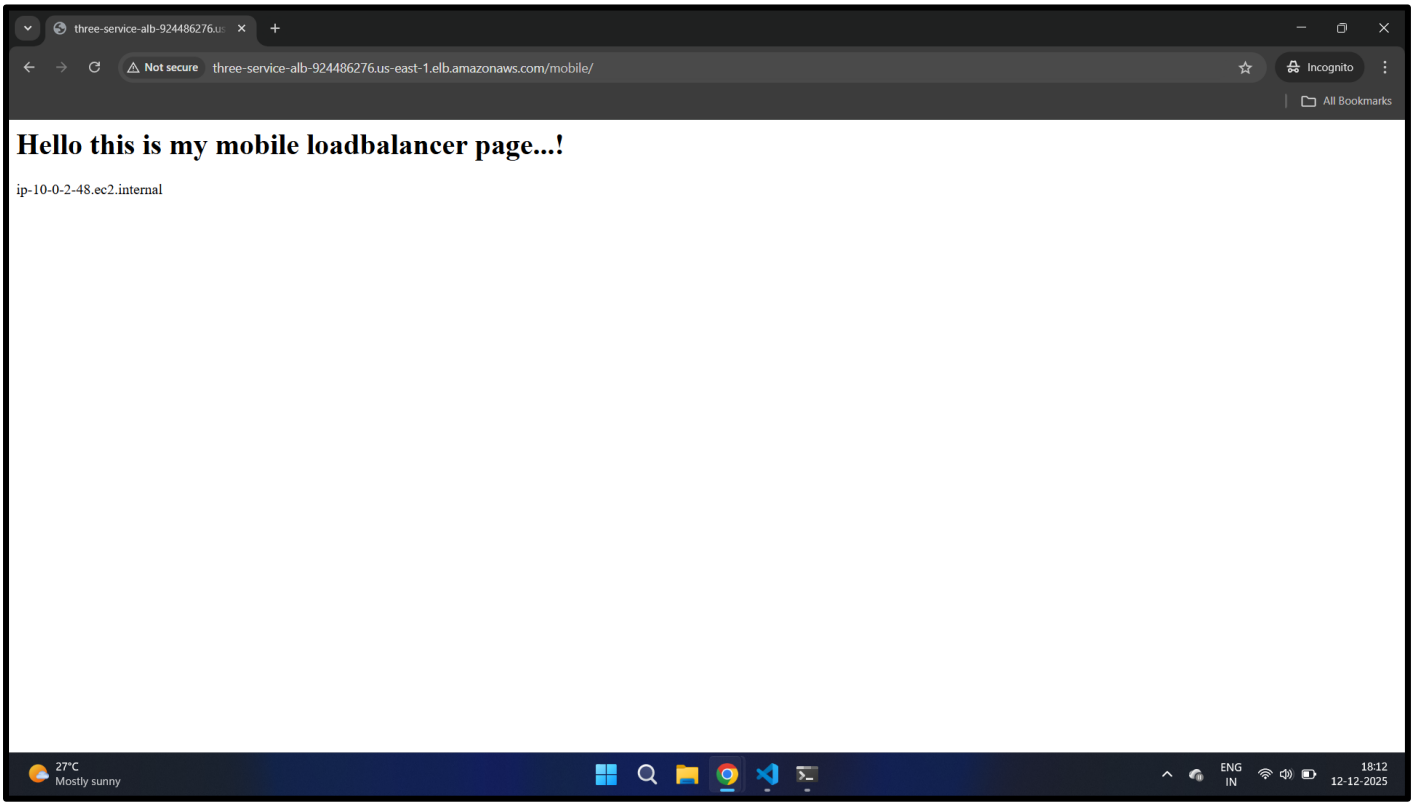
Project step-by-step-execution images.

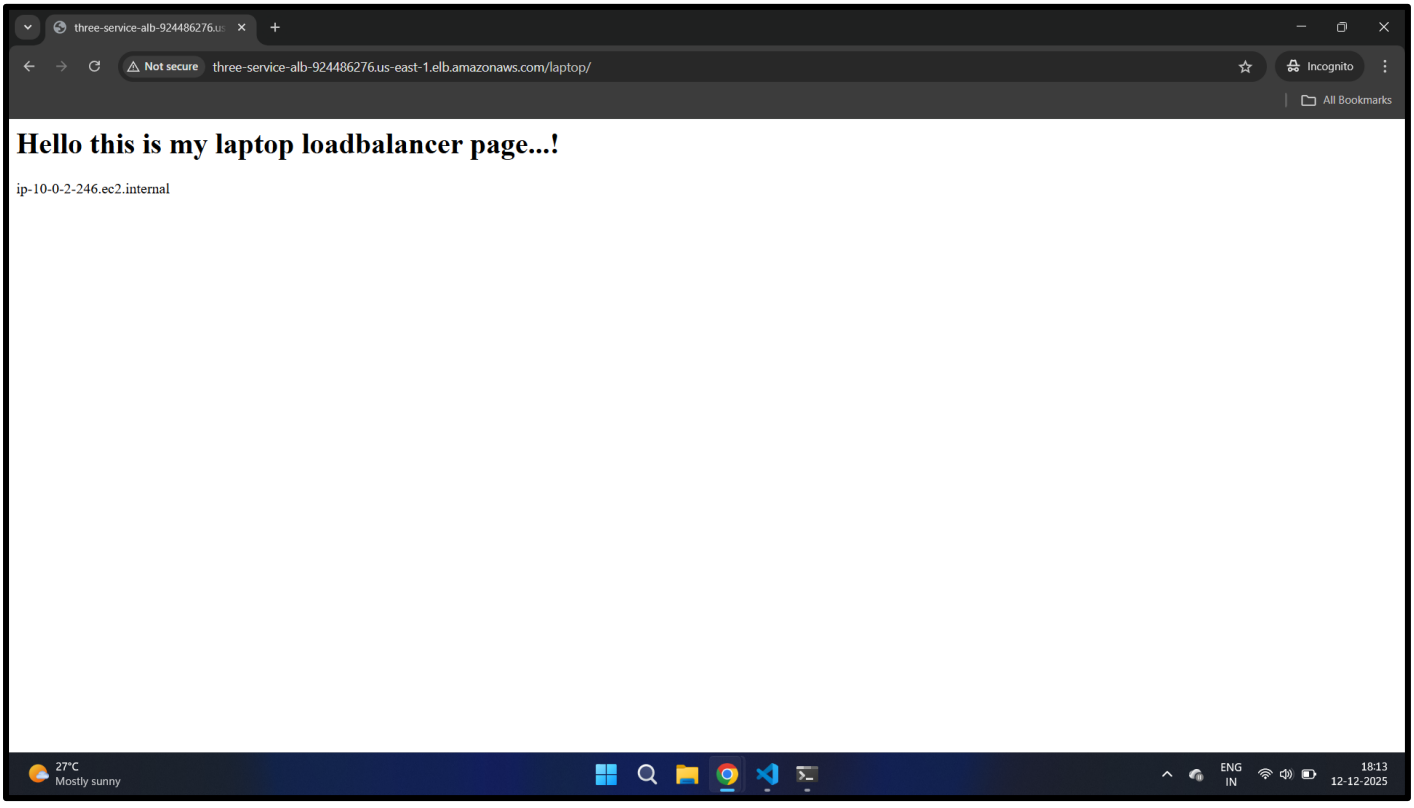


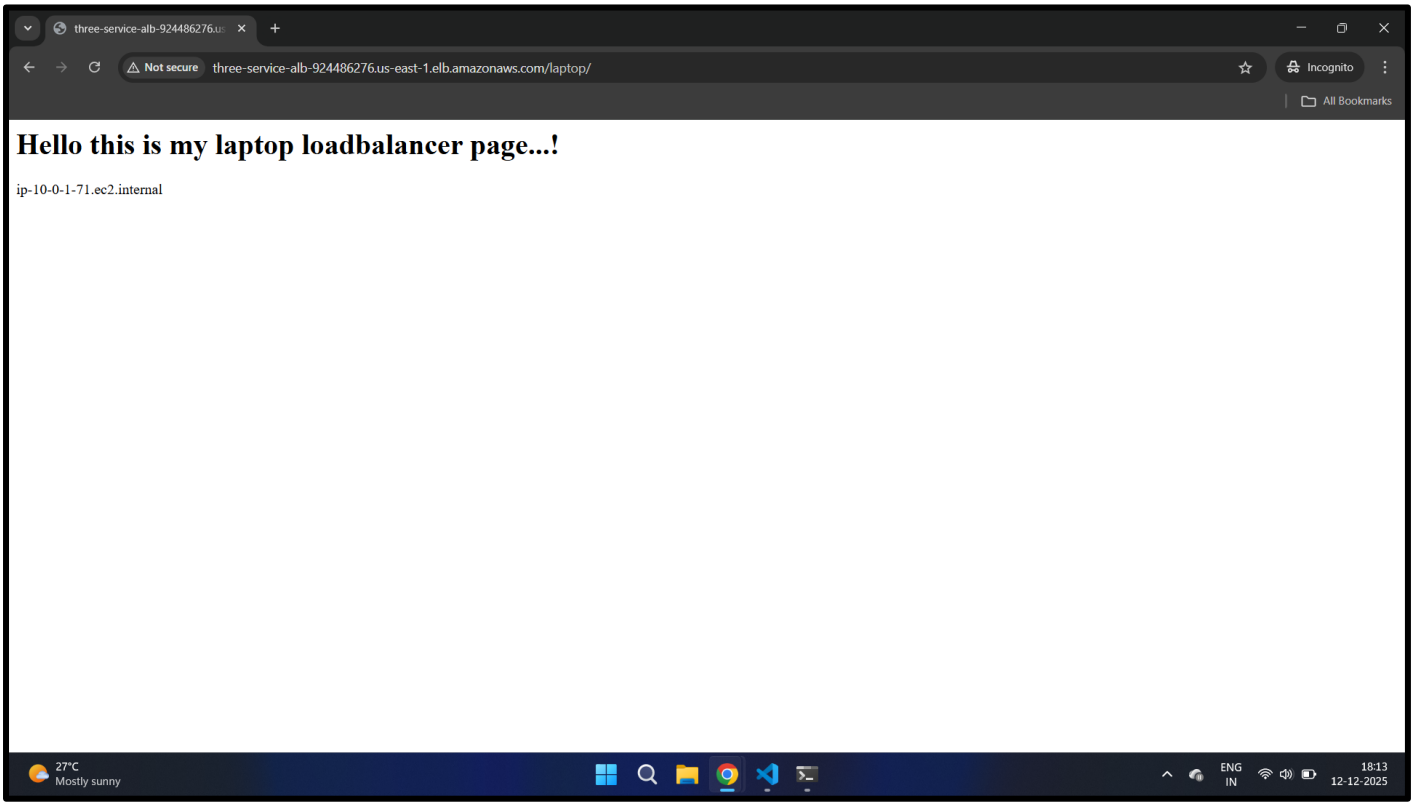


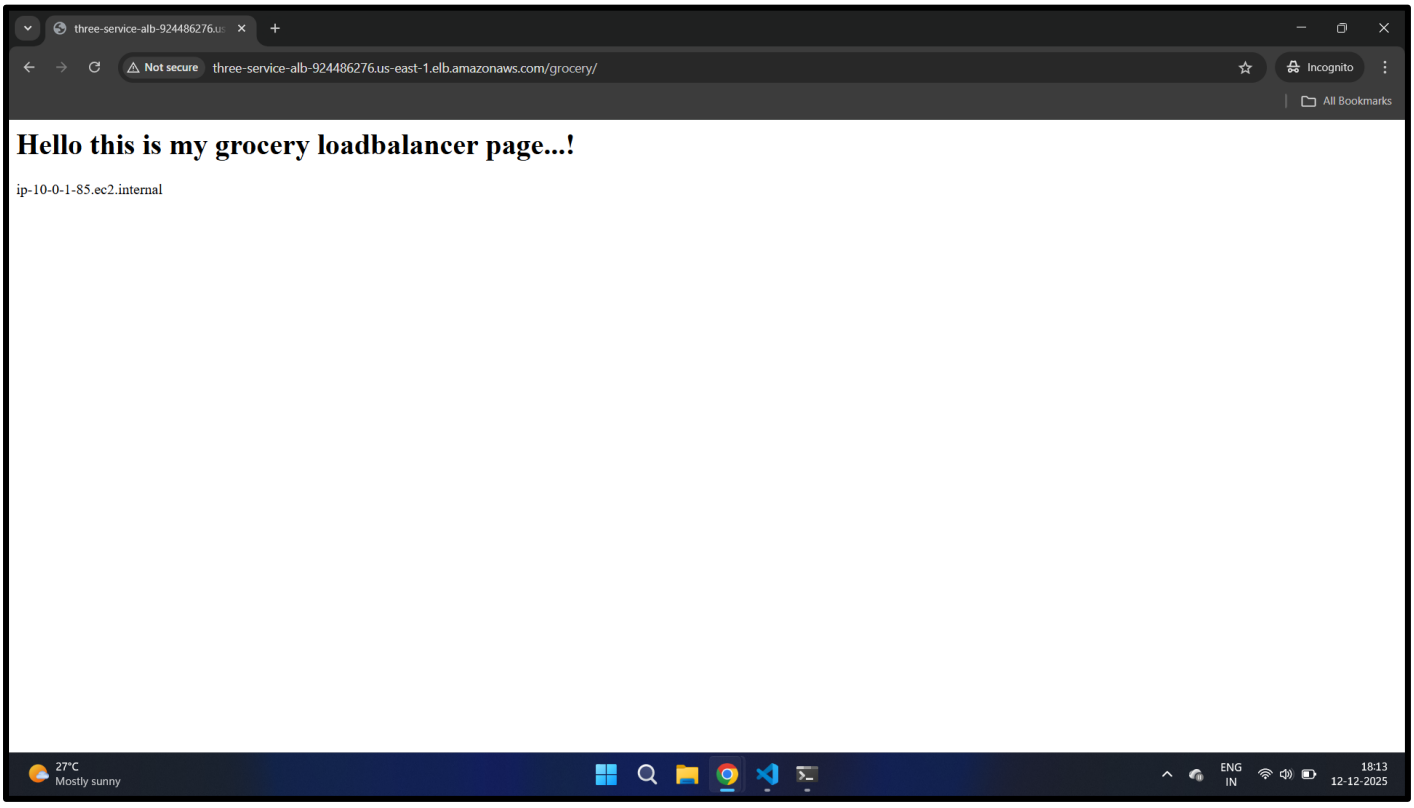

piyush_ubuntu@Piyush-Chopa
root@Piyush-Chopade: ~/terr
aws_lb.app_lb: Still creating... [00m33s elapsed]
aws_autoscaling_group.mobile_asg: Creation complete after 26s [id=mobile-asg]
aws_lb.app_lb: Still creating... [00m45s elapsed]
aws_lb.app_lb: Still creating... [00m55s elapsed]
aws_lb.app_lb: Still creating... [01m05s elapsed]
aws_lb.app_lb: Still creating... [01m15s elapsed]
aws_lb.app_lb: Still creating... [01m28s elapsed]
aws_lb.app_lb: Still creating... [01m38s elapsed]
aws_lb.app_lb: Still creating... [01m48s elapsed]
aws_lb.app_lb: Still creating... [02m01s elapsed]
aws_lb.app_lb: Still creating... [02m11s elapsed]
aws_lb.app_lb: Still creating... [02m21s elapsed]
aws_lb.app_lb: Still creating... [02m34s elapsed]
aws_lb.app_lb: Still creating... [02m44s elapsed]
aws_lb.app_lb: Still creating... [02m54s elapsed]
aws_lb.app_lb: Still creating... [03m07s elapsed]
aws_lb.app_lb: Still creating... [03m17s elapsed]
aws_lb.app_lb: Creation complete after 3m20s [id=arn:aws:elasticloadbalancing:us-east-1:148796806444:loadbalancer/app/three-service-alb/4899c8a30f994c35]
aws_lb_listener.http_listener: Creating...
aws_lb_listener.http_listener: Creation complete after 1s [id=arn:aws:elasticloadbalancing:us-east-1:148796806444:listener/app/three-service-alb/4899c8a30f994c35/75c55210ce75bb5b]
aws_lb_listener_rule.laptop_rule: Creating...
aws_lb_listener_rule.mobile_rule: Creating...
aws_lb_listener_rule.grocery_rule: Creating...
aws_lb_listener_rule.laptop_rule: Creation complete after 2s [id=arn:aws:elasticloadbalancing:us-east-1:148796806444:listener-rule/app/three-service-alb/4899c8a30f994c35/75c55210ce75bb5b/356e8e7a37a16c1f]
aws_lb_listener_rule.mobile_rule: Creation complete after 2s [id=arn:aws:elasticloadbalancing:us-east-1:148796806444:listener-rule/app/three-service-alb/4899c8a30f994c35/75c55210ce75bb5b/92d7e98db512716b]
aws_lb_listener_rule.grocery_rule: Creation complete after 2s [id=arn:aws:elasticloadbalancing:us-east-1:148796806444:listener-rule/app/three-service-alb/4899c8a30f994c35/75c55210ce75bb5b/921661f043ca9bb8]
Apply complete! Resources: 23 added, 0 changed, 0 destroyed.
Outputs:
alb_dns = "three-service-alb-1859170443.us-east-1.elb.amazonaws.com"
root@Piyush-Chopade:~/terraform#
27°C
Mostly sunny
ENG
IN
17:54
12-12-2025

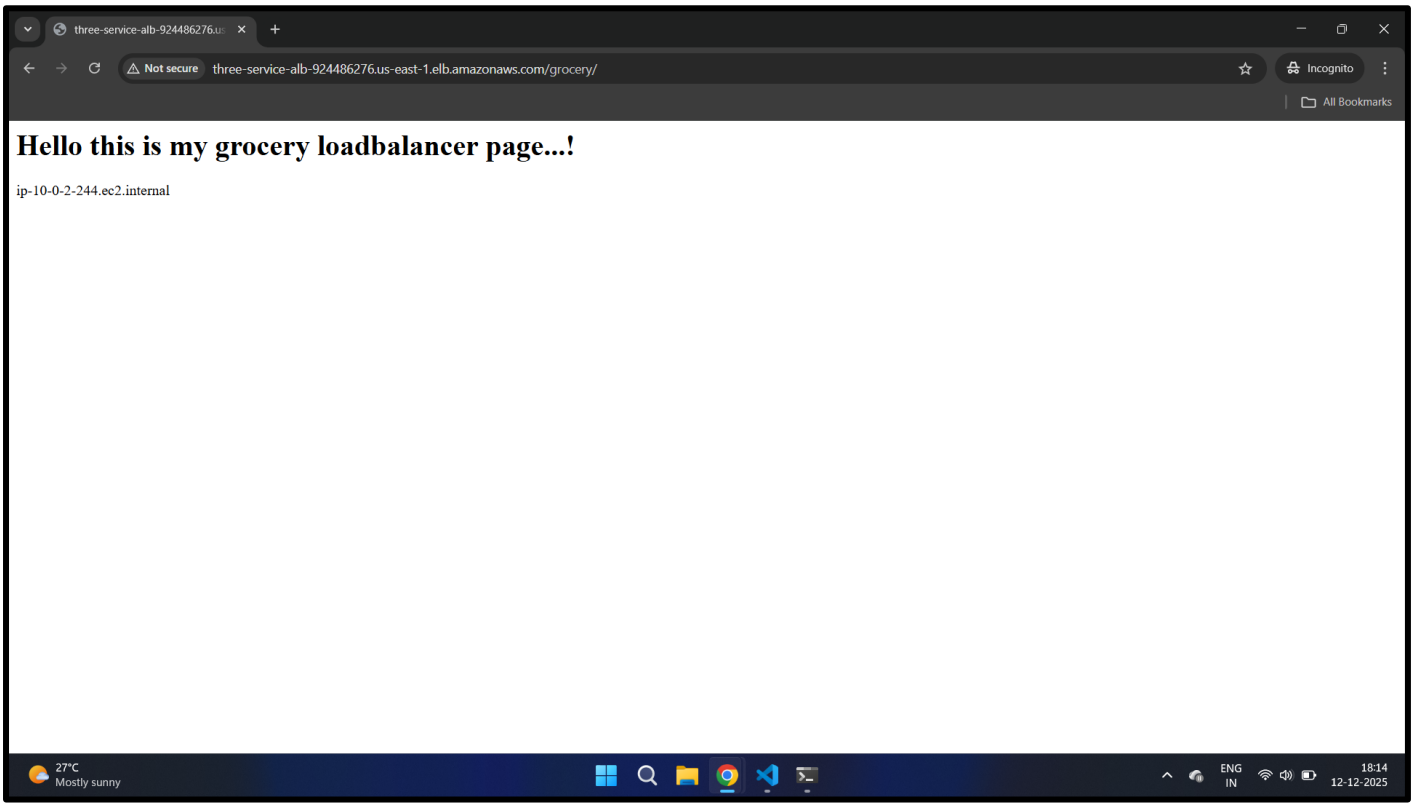

three-service-alb-924486276.us
Not secure
three-service-alb-924486276.us-east-1.elb.amazonaws.com/mobile/
Incognito
All Bookmarks
Hello this is my mobile loadbalancer page...!
ip-10-0-1-191.ec2.internal
27°C
Mostly sunny
ENG
IN
18:12
12-12-2025

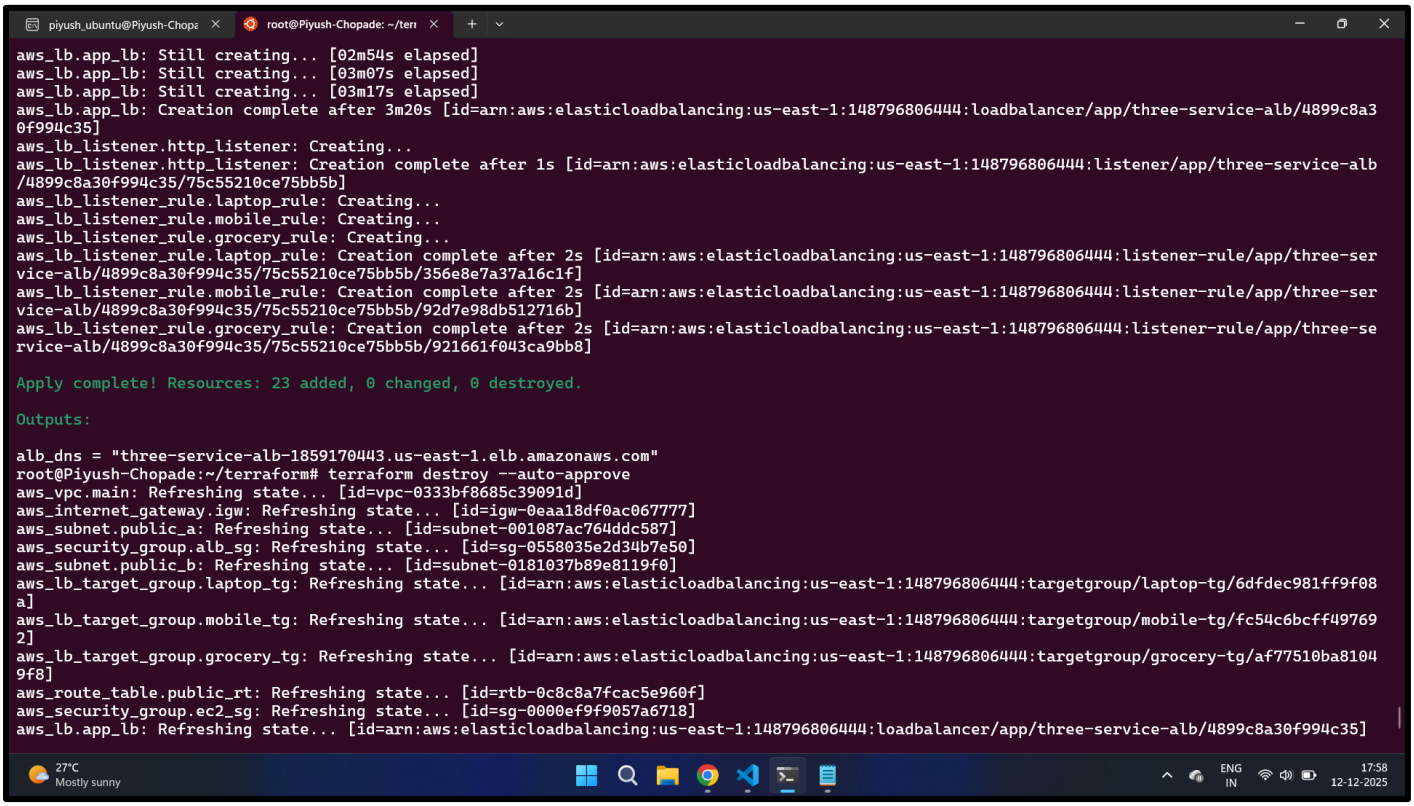

three-service-alb-924486276.us-east-1.elb.amazonaws.com/mobile/
Not secure
Incognito
All Bookmarks
Hello this is my mobile loadbalancer page...!
ip-10-0-2-48.ec2.internal
27°C
Mostly sunny
ENG
IN
18:12
12-12-2025

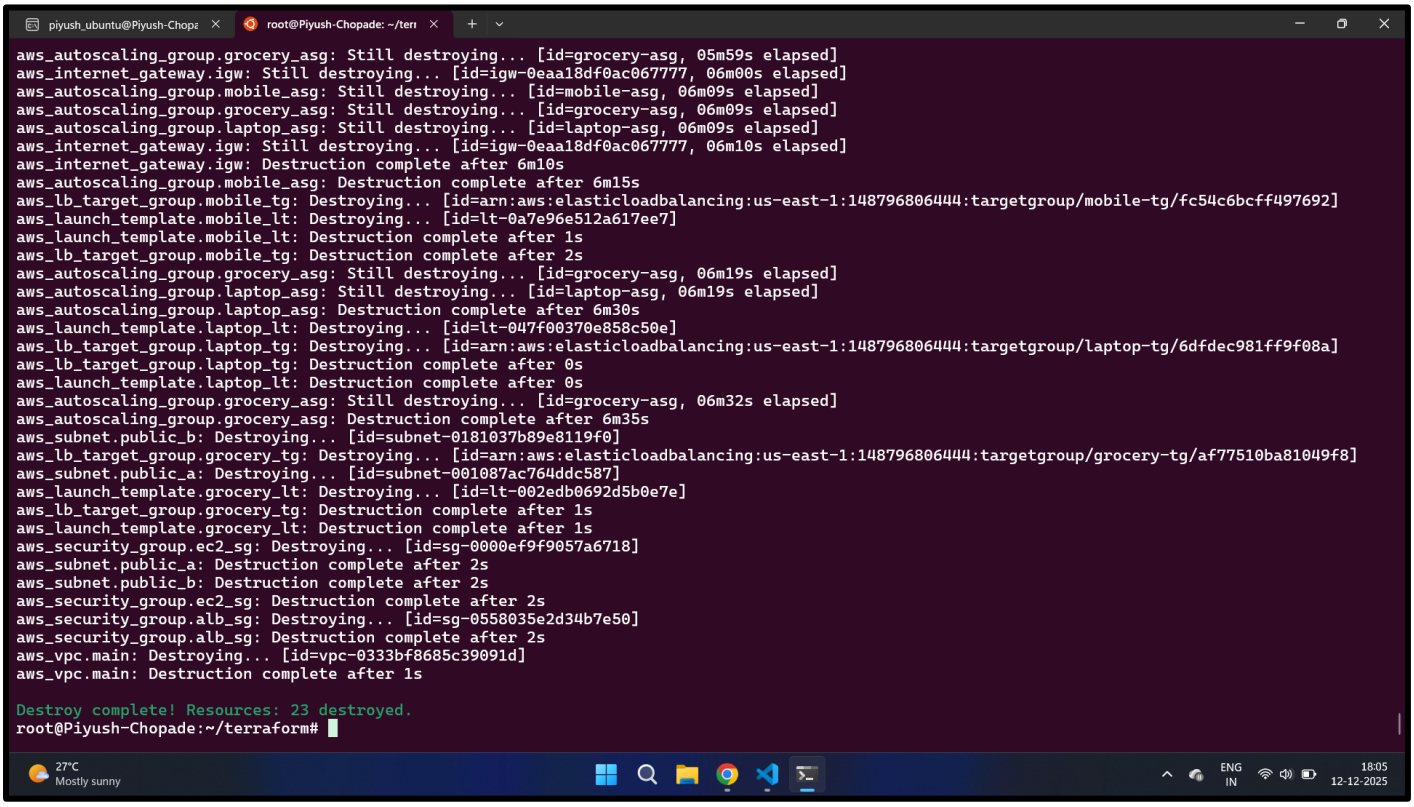

three-service-alb-924486276.us-east-1.elb.amazonaws.com/laptop/
Not secure
Incognito
All Bookmarks
Hello this is my laptop loadbalancer page...!
ip-10-0-2-246.ec2.internal
27°C
Mostly sunny
ENG
IN
18:13
12-12-2025


three-service-alb-924486276.us-east-1.elb.amazonaws.com/laptop/
Not secure
Incognito
All Bookmarks
Hello this is my laptop loadbalancer page...!
ip-10-0-1-71.ec2.internal
27°C
Mostly sunny
ENG
IN
18:13
12-12-2025


three-service-alb-924486276.us-east-1.elb.amazonaws.com/grocery/
Not secure
Incognito
All Bookmarks
Hello this is my grocery loadbalancer page...!
ip-10-0-1-85.ec2.internal
27°C
Mostly sunny
ENG
IN
18:13
12-12-2025


three-service-alb-924486276.us-east-1.elb.amazonaws.com/grocery/
Not secure
Incognito
All Bookmarks
Hello this is my grocery loadbalancer page...!
ip-10-0-2-244.ec2.internal
27°C
Mostly sunny
ENG
IN
18:14
12-12-2025


aws_lb.app_lb: Still creating... [02m54s elapsed]
aws_lb.app_lb: Still creating... [03m07s elapsed]
aws_lb.app_lb: Still creating... [03m17s elapsed]
aws_lb.app_lb: Creation complete after 3m20s [id=arn:aws:elasticloadbalancing:us-east-1:148796806444:loadbalancer/app/three-service-alb/4899c8a30f994c35]
aws_lb_listener.http_listener: Creating...
aws_lb_listener.http_listener: Creation complete after 1s [id=arn:aws:elasticloadbalancing:us-east-1:148796806444:listener/app/three-service-alb/4899c8a30f994c35/75c55210ce75bb5b]
aws_lb_listener_rule.laptop_rule: Creating...
aws_lb_listener_rule.mobile_rule: Creating...
aws_lb_listener_rule.grocery_rule: Creating...
aws_lb_listener_rule.laptop_rule: Creation complete after 2s [id=arn:aws:elasticloadbalancing:us-east-1:148796806444:listener-rule/app/three-service-alb/4899c8a30f994c35/75c55210ce75bb5b/356e8e7a37a16c1f]
aws_lb_listener_rule.mobile_rule: Creation complete after 2s [id=arn:aws:elasticloadbalancing:us-east-1:148796806444:listener-rule/app/three-service-alb/4899c8a30f994c35/75c55210ce75bb5b/92d7e98db512716b]
aws_lb_listener_rule.grocery_rule: Creation complete after 2s [id=arn:aws:elasticloadbalancing:us-east-1:148796806444:listener-rule/app/three-service-alb/4899c8a30f994c35/75c55210ce75bb5b/921661f043ca9bb8]
Apply complete! Resources: 23 added, 0 changed, 0 destroyed.
Outputs:
alb_dns = "three-service-alb-1859170443.us-east-1.elb.amazonaws.com"
root@Piyush-Chopade:~/terraform# terraform destroy --auto-approve
aws_vpc.main: Refreshing state... [id=vpc-0333bf8685c39091d]
aws_internet_gateway.igw: Refreshing state... [id=igw-0eaa18df0ac067777]
aws_subnet.public_a: Refreshing state... [id=subnet-001087ac764ddc587]
aws_security_group.alb_sg: Refreshing state... [id=sg-0558035e2d34b7e50]
aws_subnet.public_b: Refreshing state... [id=subnet-0181037b89e8119f0]
aws_lb_target_group.laptop_tg: Refreshing state... [id=arn:aws:elasticloadbalancing:us-east-1:148796806444:targetgroup/laptop-tg/6dfdec981ff9f08a]
aws_lb_target_group.mobile_tg: Refreshing state... [id=arn:aws:elasticloadbalancing:us-east-1:148796806444:targetgroup/mobile-tg/fc54c6bcff497692]
aws_lb_target_group.grocery_tg: Refreshing state... [id=arn:aws:elasticloadbalancing:us-east-1:148796806444:targetgroup/grocery-tg/af77510ba81049f8]
aws_route_table.public_rt: Refreshing state... [id=rtb-0c8c8a7fcac5e960f]
aws_security_group.ec2_sg: Refreshing state... [id=sg-0000ef9f9057a6718]
aws_lb.app_lb: Refreshing state... [id=arn:aws:elasticloadbalancing:us-east-1:148796806444:loadbalancer/app/three-service-alb/4899c8a30f994c35]
27°C
Mostly sunny
ENG
IN
17:58
12-12-2025


piyush_ubuntu@Piyush-Chopa
root@Piyush-Chopade: ~/terr
aws_autoscaling_group.grocery_asg: Still destroying... [id=grocery-asg, 05m59s elapsed]
aws_internet_gateway.igw: Still destroying... [id=igw-0eaa18df0ac067777, 06m00s elapsed]
aws_autoscaling_group.mobile_asg: Still destroying... [id=mobile-asg, 06m09s elapsed]
aws_autoscaling_group.grocery_asg: Still destroying... [id=grocery-asg, 06m09s elapsed]
aws_autoscaling_group.laptop_asg: Still destroying... [id=laptop-asg, 06m09s elapsed]
aws_internet_gateway.igw: Still destroying... [id=igw-0eaa18df0ac067777, 06m10s elapsed]
aws_internet_gateway.igw: Destruction complete after 6m10s
aws_autoscaling_group.mobile_asg: Destruction complete after 6m15s
aws_lb_target_group.mobile_tg: Destroying... [id=arn:aws:elasticloadbalancing:us-east-1:148796806444:targetgroup/mobile-tg/fc54c6bcff497692]
aws_launch_template.mobile_lt: Destroying... [id=lt-0a7e96e512a617ee7]
aws_launch_template.mobile_lt: Destruction complete after 1s
aws_lb_target_group.mobile_tg: Destruction complete after 2s
aws_autoscaling_group.grocery_asg: Still destroying... [id=grocery-asg, 06m19s elapsed]
aws_autoscaling_group.laptop_asg: Still destroying... [id=laptop-asg, 06m19s elapsed]
aws_autoscaling_group.laptop_asg: Destruction complete after 6m30s
aws_launch_template.laptop_lt: Destroying... [id=lt-047f00370e858c50e]
aws_lb_target_group.laptop_tg: Destroying... [id=arn:aws:elasticloadbalancing:us-east-1:148796806444:targetgroup/laptop-tg/6dfdec981ff9f08a]
aws_lb_target_group.laptop_tg: Destruction complete after 0s
aws_launch_template.laptop_lt: Destruction complete after 0s
aws_autoscaling_group.grocery_asg: Still destroying... [id=grocery-asg, 06m32s elapsed]
aws_autoscaling_group.grocery_asg: Destruction complete after 6m35s
aws_subnet.public_b: Destroying... [id=subnet-0181037b89e8119f0]
aws_lb_target_group.grocery_tg: Destroying... [id=arn:aws:elasticloadbalancing:us-east-1:148796806444:targetgroup/grocery-tg/af77510ba81049f8]
aws_subnet.public_a: Destroying... [id=subnet-001087ac764ddc587]
aws_launch_template.grocery_lt: Destroying... [id=lt-002edb0692d5b0e7e]
aws_lb_target_group.grocery_tg: Destruction complete after 1s
aws_launch_template.grocery_lt: Destruction complete after 1s
aws_security_group.ec2_sg: Destroying... [id=sg-0000ef9f9057a6718]
aws_subnet.public_a: Destruction complete after 2s
aws_subnet.public_b: Destruction complete after 2s
aws_security_group.ec2_sg: Destruction complete after 2s
aws_security_group.alb_sg: Destroying... [id=sg-0558035e2d34b7e50]
aws_security_group.alb_sg: Destruction complete after 2s
aws_vpc.main: Destroying... [id=vpc-0333bf8685c39091d]
aws_vpc.main: Destruction complete after 1s
Destroy complete! Resources: 23 destroyed.
root@Piyush-Chopade:~/terraform#
27°C
Mostly sunny
ENG
IN
18:05
12-12-2025